

Quiz 5 Reflection

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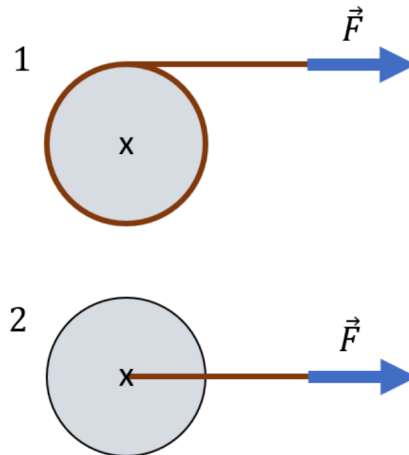
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1. Collision Ball and Box 1

1.1. Specify Question

Question 4: Hockey Physicist



Two identical pucks are laying flat on an icy surface and both are pulled via a thin string by a hockey player (who is a physicist at heart). The hockey player pulls with the same constant horizontal force F on both pucks for 2 seconds. In the first scenario, the thin string is wound around the outside of the puck, while in the second it is attached to its center.

a) Which puck achieves a higher speed?

Puck #2

✗ 0%

b) In which scenario does the hockey player do more work as they pull on the string?

Scenario #2

✗ 0%

This question is complete and cannot be answered again.

Correct answer

Same for both Scenario #1

1.2. Diagnosis Phase

I mistakenly thought about it from an energy perspective. Words: 9

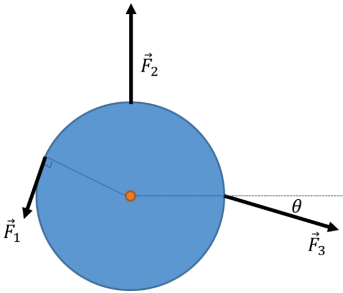
1.3. Correction Phase

Words: 0

2. Angular Velocity of Disk

2.1. Specify Question

Question 5: Angular Velocity of Disk



A solid uniform disk with mass $M = 10\text{ kg}$ and radius $R = 30\text{ cm}$ is attached to a fixed axle going through its center of mass. It is initially at rest when the following forces $F_1 = 16\text{ N}$, $F_2 = 20\text{ N}$ and $F_3 = 32\text{ N}$ are applied to it. If $\theta = 30^\circ$, what is the disk's angular velocity after 2 s (in rad/s)?

Assume that "out of the page" is positive and give your answer to 2 decimal places.

$\omega =$

42.666 rad / s

?

✖

0%

This question is complete and cannot be answered again.

Correct answer

$\omega = 0.0\text{ radian / second}$

2.2. Diagnosis Phase

I misread the diagram and mistakenly thought that the $\vec{F}_3$ was pointing up instead of down so when I went to calculate velocity I got:

$$\tau = \vec{F} \times R$$
$$\tau_1 = \vec{F}_1 R \sin(90^\circ) = (16\text{N})(0.3\text{m})(1) = 4.8\text{N}$$
$$\tau_2 = \vec{F}_3 R \sin(180^\circ) = (20\text{N})(0.3\text{m})(0) = 0\text{ N/m}$$

I incorrectly thought the $\vec{F}_3$ was pointing up, so I calculated its torque on the axis

$$\tau_3 = \vec{F}_2 R \sin(60^\circ) = (32\text{N})(0.3\text{m})(0.5) = 8.352\text{ N/m}$$
$$\tau_{\text{total}} = \tau_1 + \tau_2 + \tau_3 = ()$$

Words: 39

2.3. Correction Phase

Interpreting the direction of $\vec{F}_3$ correctly.

$$\tau = \vec{F} \times R$$
$$\tau_1 = \vec{F}_1 R \sin(90^\circ) = (16\text{N})(32\text{ cm})(1) = 0$$

$90^\circ + \theta = 120^\circ$ “ is the angle between radius and force

$$\tau_2 = \vec{F}_2 R \sin(120^\circ) = (20\text{N})(32\text{ cm})$$
$$\tau_3 = \vec{F}_1 R \sin(90^\circ)$$
$$\tau_{\text{total}} = \tau_1 + \tau_2 + \tau_3 = ()$$

Words: 12